EE 66 Signals, Dynamics, and Information
Fall 2026 Note 7

7.1 Spectral Theorem

So far, we have explored the usefulness of diagonal and diagonalizable matrices. Another useful type of matrix is the **symmetric matrix**—matrices that are equal to their transpose. In this note, we will discuss the Spectral Theorem, which provides us with useful properties of symmetric matrices, particularly when it comes to diagonalization.

Definition 7.1 (Spectral Theorem for Real Symmetric Matrices):

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be real and symmetric. Then:

(i) The eigenvalues of $\mathbf{A}$ are real.

(ii) There is an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A}$.

The conclusion of these facts: $\mathbf{A}$ may be *orthonormally diagonalized*: $\mathbf{A} = \mathbf{V}\Lambda\mathbf{V}^\top$ where $\mathbf{V} \in \mathbb{R}^{n \times n}$ is an orthonormal matrix of eigenvectors of $\mathbf{A}$, and $\Lambda \in \mathbb{R}^{n \times n}$ is a real diagonal matrix of eigenvalues.

Orthonormal diagonalization is a key ingredient in proofs and algorithms involving symmetric matrices.

7.1.1 Proof that Symmetric Matrices Have Real Eigenvalues

Take an arbitrary eigenvalue λ of $\mathbf{A}$ with corresponding eigenvector $\mathbf{v}$. Then

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}. \tag{1}$$

Taking the conjugate of both sides and using the fact that $\mathbf{A}$ is real ($\mathbf{A} = \mathbf{A}^*$), we get

$$\mathbf{A}\mathbf{v}^* = \mathbf{A}^*\mathbf{v}^* = \mathbf{A}\mathbf{v}^* = \lambda\mathbf{v}^* = \lambda^*\mathbf{v}^*. \tag{2}$$

Taking advantage of the fact that $\mathbf{A}$ is symmetric ($\mathbf{A} = \mathbf{A}^\top$), we take the transpose of both sides to get

$$\mathbf{A}\mathbf{v}^* = \lambda^*\mathbf{v}^* \tag{3}$$

$$(\mathbf{v}^*)^\top\mathbf{A}^\top = \lambda^*(\mathbf{v}^*)^\top \tag{4}$$

$$(\mathbf{v}^*)^\top\mathbf{A} = \lambda^*(\mathbf{v}^*)^\top. \tag{5}$$

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Then we multiply by $\mathbf{v}$ on both sides to get

$$(\mathbf{v}^*)^\top \mathbf{A}\mathbf{v} = \lambda^* (\mathbf{v}^*)^\top \mathbf{v} \tag{6}$$
$$\lambda (\mathbf{v}^*)^\top \mathbf{v} = \lambda^* (\mathbf{v}^*)^\top \mathbf{v} \tag{7}$$
$$0 = (\lambda - \lambda^*)(\mathbf{v}^*)^\top \mathbf{v} \tag{8}$$

using the fact that $(\mathbf{v}, \lambda)$ is an eigenvector-eigenvalue pair of $\mathbf{A}$. Now

$$(\mathbf{v}^*)^\top \mathbf{v} = \sum_{i=1}^{n} v_i^* \cdot v_i = \sum_{i=1}^{n} |v_i|^2. \tag{9}$$

so $(\mathbf{v}^*)^\top \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$. Since $\mathbf{v}$ is an eigenvector, which means it's nonzero, we know that $(\mathbf{v}^*)^\top \mathbf{v}$ is nonzero, and thus that $\lambda - \lambda^* = 0$. Therefore, $\lambda = \lambda^*$ so λ is real. Since λ is an arbitrary eigenvalue, all eigenvalues of $\mathbf{A}$ are real.

7.1.2 Proof that Symmetric Matrices Have Orthogonal Eigenvectors

The interplay between symmetric matrices and orthogonal eigenvectors is a cornerstone in linear algebra. Firstly, a symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is guaranteed to have orthogonal eigenvectors. Let us try to prove this fact.

Consider a symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, and let λ_1 and λ_2 be distinct eigenvalues of $\mathbf{A}$ with corresponding eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively. We aim to show that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal.

From the definition of eigenvectors and eigenvalues, we have:

$$\mathbf{A}\mathbf{v}_1 = \lambda_1 \mathbf{v}_1,$$
$$\mathbf{A}\mathbf{v}_2 = \lambda_2 \mathbf{v}_2$$

Multiplying both sides of the first equation by $\mathbf{v}_2^\top$ and both sides of the second by $\mathbf{v}_1^\top$ gives:

$$\mathbf{v}_2^\top \mathbf{A}\mathbf{v}_1 = \lambda_1 \mathbf{v}_2^\top \mathbf{v}_1,$$
$$\mathbf{v}_1^\top \mathbf{A}\mathbf{v}_2 = \lambda_2 \mathbf{v}_1^\top \mathbf{v}_2$$

Notice that this is a scalar so: $\mathbf{v}_1^\top \mathbf{A}\mathbf{v}_2 = (\mathbf{v}_1^\top \mathbf{A}\mathbf{v}_2)^\top = \mathbf{v}_2^\top \mathbf{A}^\top \mathbf{v}_1 = \mathbf{v}_2^\top \mathbf{A}\mathbf{v}_1$. The last step is true because $\mathbf{A}$ is symmetric, hence $\mathbf{A} = \mathbf{A}^\top$. Therefore, we can equate the right-hand sides of the above equations:

$$\lambda_1 \mathbf{v}_2^\top \mathbf{v}_1 = \lambda_2 \mathbf{v}_1^\top \mathbf{v}_2$$
$$\lambda_1 \mathbf{v}_1^\top \mathbf{v}_2 = \lambda_2 \mathbf{v}_1^\top \mathbf{v}_2$$
$$(\lambda_1 - \lambda_2)\mathbf{v}_1^\top \mathbf{v}_2 = 0$$

Note that we use the fact that $\mathbf{v}_1^\top \mathbf{v}_2 = \mathbf{v}_2^\top \mathbf{v}_1$.

Given $\lambda_1 \neq \lambda_2$, it follows that $\mathbf{v}_1^\top \mathbf{v}_2 = 0$, demonstrating that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal.

If we have that each eigenvalue of $\mathbf{A}$ is distinct, then we have n orthonormal eigenvectors that form a basis for $\mathbb{R}^n$ as desired. Note that the eigenvectors can be orthonormal since scaling eigenvectors gives us an eigenvector.

For the general case where we have repeated eigenvalues, this is still true. However the proof of this statement is out of scope for the class, requiring a lot more linear algebra terminology and is quite technical. If you are interested in the full proof, *Linear Algebra* by Friedberg et. al. has a chapter that discusses the full statement.

Caution: The above proofs and statements only apply for symmetric matrices; this is NOT true for all matrices in general. Make sure you know the matrix you are dealing with is symmetric before you assume it can be orthonormally diagonalized with real eigenvalues.

7.1.3 Proof that Orthogonal Eigenvectors Imply a Symmetric Matrix

Assume a matrix $\mathbf{A}$ has a set of orthogonal eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ with corresponding eigenvalues $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$. Let $\mathbf{Q}$ be the matrix with columns as the normalized eigenvectors of $\mathbf{A}$ and Λ be the diagonal matrix with the eigenvalues of $\mathbf{A}$.

Since the eigenvectors are orthogonal and normalized, $\mathbf{Q}$ is an orthogonal matrix, and hence $\mathbf{Q}^\top \mathbf{Q} = \mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix.

We can write $\mathbf{A}$ as:

$$\mathbf{A} = \mathbf{Q}\Lambda\mathbf{Q}^\top.$$

To show $\mathbf{A}$ is symmetric, we need to prove $\mathbf{A}^\top = \mathbf{A}$. Taking the transpose of the above equation:

$$\mathbf{A}^\top = (\mathbf{Q}\Lambda\mathbf{Q}^\top)^\top = \mathbf{Q}\Lambda^\top\mathbf{Q}^\top.$$

Since Λ is diagonal, $\Lambda^\top = \Lambda$. Thus, we obtain:

$$\mathbf{A}^\top = \mathbf{Q}\Lambda\mathbf{Q}^\top = \mathbf{A},$$

which shows that $\mathbf{A}$ is symmetric.

7.1.4 Outer Product Form of Spectral Theorem

We usually write the diagonalized form of a matrix using matrix multiplication:

$$\mathbf{A} = \mathbf{V}\Lambda\mathbf{V}^{-1} \tag{10}$$

If $\mathbf{A}$ is symmetric, we now know from the spectral theorem that:

$$\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top \tag{11}$$

The key difference here is the presence of $\mathbf{V}^\top$, which uses the same subvectors as $\mathbf{V}$ (unlike $\mathbf{V}^{-1}$, whose subvectors have a more complex relation to $\mathbf{V}$):

$$\mathbf{V} = \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix} \tag{12}$$

$$\mathbf{V}^\top = \begin{bmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_n^\top \end{bmatrix} \tag{13}$$

Thus, for symmetric matrices, we can rewrite their diagonal decomposition in terms of the eigenvectors $\mathbf{v}_i$:

$$\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top = \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_n^\top \end{bmatrix} = \sum_{i=1}^{n} \lambda_i \mathbf{v}_i \mathbf{v}_i^\top \tag{14}$$

Notice that $\mathbf{v}_i\mathbf{v}_i^\top$ are examples of outer products. Thus, the conclusion here is that a symmetric matrix $\mathbf{A}$ can be written as the sum of outer products of its eigenvectors, each scaled by the associated eigenvalue.

This is also why the Spectral Theorem is called "Spectral" Theorem; the "spectrum" in this case refers to the set of eigenvalues of a matrix, and the outer products are the associated subcomponents of the overall matrix. Thus, the Spectral Theorem is decomposing a matrix into its spectral components.